

Reaction Stoichiometry

Chemical reactions occur between components in the aqueous phase and between minerals and aqueous components. Let R_{aq} be the number of reactions between aqueous components and R_{mn} be the number of reactions between minerals and aqueous components. The components in the aqueous phase comprise gaseous components that are soluble in the aqueous phase (n_c components) as well as components that exist only in the aqueous phase (n_a components). Let n_m be the number of mineral components, n_{aq} be the total number of components in the aqueous phase ($n_{aq} = n_c + n_a$) and n_{ct} be the total number of components ($n_{ct} = n_c + n_a + n_m$).

A reaction between species in the aqueous phase has the following stoichiometry:

$$\sum_{i=1}^{n_{aq}} v_{ij} A_i = 0, \quad j = 1, \dots, R_{aq} \quad (D5.1)$$

where A_i is the chemical symbol for the i -th aqueous species. The dissolution or precipitation chemical reaction for a mineral has the following stoichiometry:

$$\sum_{i=1}^{n_{ct}} v_{ij} A_i = 0, \quad j = 1, \dots, R_{mn} \quad (D5.2)$$

An example of the reaction Equation (D5.3) is the dissociation of $\text{CO}_2(aq)$, i.e.



An example of the reaction Equation (D5.4) is dissolution of Calcite (CaCO_3):



The chemical reactions Equation (D5.1) are homogeneous reactions that involve only components in the aqueous phase. They do not involve any minerals and therefore the summation involves only n_{aq} components. The chemical reactions in Equation (D5.2) are heterogeneous reactions that involve mineral species and aqueous species. Normally, a mineral reacts only with aqueous species and not with other minerals. Therefore, the stoichiometric coefficients associated with mineral species in Equation (D5.2) are all zero except for the one corresponding to the reacting mineral. Also, in this paper, all mineral reactions are written as dissolution reactions with the stoichiometric coefficient for the reacting mineral equal to -1. There is no loss of generality by adopting this convention, since precipitation can also be modeled as discussed later.

Reactions between components in the aqueous phase are fast relative to mineral dissolution/precipitation reactions. Therefore, intra-aqueous reactions are represented as chemical-equilibrium reactions whereas mineral dissolutions/precipitations are represented as rate-dependent reactions.